

A SAT-Modulo-Symmetries verification of the Erdős–Gyárfás power-of-two cycle conjecture for minimum-degree-3 graphs up to 30 vertices

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Abstract

The Erdős–Gyárfás conjecture (1995) asserts that every graph with minimum degree at least 3 contains a cycle whose length is a power of two. The conjecture is open; prior computer search established that any counterexample on the *general* minimum-degree-3 class has at least 17 vertices (Royle and Markström), and that any *cubic* (3-regular) counterexample has at least 30 vertices (Markström, 2004). We give, to our knowledge, the first application of SAT-based methods to this conjecture. Using SAT Modulo Symmetries (SMS) with the Glasgow subgraph solver as a complete forbidden-subgraph propagator, we verify that *every* minimum-degree-3 graph on at most 30 vertices contains a cycle of length 4, 8, or 16. Consequently any general minimum-degree-3 counterexample must have at least 31 vertices, improving the two-decade-old general bound from 17 to 31; since the cubic class is a subclass, this also improves the cubic bound from 30 to 31. We corroborate the result with an independent ground-truth check at $n = 10$, reproduction of the $n \leq 16$ baseline, agreement with an independent CEGAR-SAT solver for $n \leq 19$, robustness to two cardinality encodings and a second symmetry-breaking method, and positive controls; we discuss the path to a formally machine-checked proof certificate.

1 Introduction

In 1995 Erdős and Gyárfás posed the following conjecture (see Erdős [1]).

Conjecture 1 (Erdős–Gyárfás). *Every graph with minimum degree at least 3 contains a simple cycle whose length is a power of two.*

Here “power of two” means 2^k for some integer $k \geq 2$, i.e. a cycle of length 4, 8, 16, $\dots$. The conjecture remains open; Erdős offered \$100 for a proof and \$50 for a counterexample [2]. It has been confirmed for several restricted classes, including 3-connected cubic planar graphs [4], P_8 -free graphs [7], and P_{10} -free graphs [8]; the most recent structural progress shows that any *minimal* counterexample is “predominantly cubic” [10].

On the computational side, two distinct frontiers are known, and the distinction is central to the present paper. Write $\delta(G)$ for the minimum degree of G .

- **General** ($\delta \geq 3$, arbitrary maximum degree): computer searches of Royle and Markström are commonly cited as establishing that any counterexample has at least 17 vertices, i.e. Conjecture 1 holds for all $\delta \geq 3$ graphs on at most 16 vertices [16].

*Preprint draft; author and affiliation to be finalized. This is a computational result; see §4 for the verification and §5 for scope.

- **Cubic** (3-regular): Markström [3] verified all cubic graphs on at most 29 vertices, so any cubic counterexample has at least 30 vertices.

A targeted, adversarial literature search (covering theses, technical reports, OEIS and House of Graphs, and the two most recent surveys of computer-assisted graph theory and of SAT Modulo Symmetries [12, 13]) found no improvement of the *general* bound past 17 after 2004, and no prior application of SAT, SAT Modulo Symmetries, constraint programming, or isomorph-free generation to Conjecture 1.

Contribution. We give the first SAT-based attack on Conjecture 1. Using SAT Modulo Symmetries (SMS) [11, 12] together with the Glasgow subgraph solver [14], we verify Conjecture 1 for *all* minimum-degree-3 graphs on n vertices for every n up to 30:

Theorem 1. *Every graph with minimum degree at least 3 on at most 30 vertices contains a cycle of length 4, 8, or 16. Consequently, any minimum-degree-3 counterexample to the Erdős–Gyárfás conjecture has at least 31 vertices.*

This improves the general bound from 17 to 31, a +14-vertex extension over a search space that — unlike the cubic case — includes all non-regular graphs of minimum degree 3.

General versus cubic (a point to read carefully). Our verified order, 30, coincides numerically with Markström’s cubic *bound* of 30, but the two statements are different and should not be conflated. Markström verified the *cubic* class up to order 29 (cubic bound ≥ 30). We verify the *entire* minimum-degree-3 class — a strictly larger family, dominated by non-regular graphs — up to order 30 (general bound ≥ 31). Since the cubic class is contained in the minimum-degree-3 class, our search in particular verifies all cubic graphs on ≤ 30 vertices, which incidentally also improves the cubic bound from 30 to 31. Thus the genuinely new content is twofold: the *non-cubic* minimum-degree-3 graphs on $17 \leq n \leq 30$ vertices, and the methodology (SAT/SMS), which had not previously been applied to this conjecture.

Related work. The conjecture is known for several restricted classes: $K_{1,m}$ -free graphs [5], planar claw-free graphs [6], 3-connected cubic planar graphs [4], and the P_t -free families P_8 [7], P_{10} [8], and (with the aid of a computer search) P_{13} [9]. These are theorems for infinite hereditary classes and do not bound the order of a general counterexample; in particular, the computer search of [9] assists a structural proof for the P_{13} -free class and is unrelated to the exhaustive, isomorph-free SAT enumeration used here. The most recent structural progress, Carr [10], shows that at least $4/7$ of the vertices of any minimal counterexample have degree exactly 3; it is an arXiv preprint that uses no computation. Our methodological ingredients — SAT Modulo Symmetries [11, 12] and the Glasgow subgraph solver [14] — have been applied to other graph-existence questions (e.g. the 3-Decomposition Conjecture on cubic graphs), but, as noted above, not to Conjecture 1.

2 Method

For a fixed number of vertices n we decide the following question with a single SMS call: *does there exist a graph G on n vertices with $\delta(G) \geq 3$ that contains no cycle of length 4, 8, or 16?* The reduction to these three cycle lengths is justified by the following elementary observation.

Proposition 1. *For $3 \leq n \leq 31$, a graph on n vertices contains a cycle whose length is a power of two if and only if it contains C_4 , C_8 , or C_{16} as a (non-induced) subgraph.*

Proof. A cycle of length ℓ is precisely a C_ℓ subgraph. The powers of two that are valid cycle lengths (at least 3) and at most $n \leq 31$ are exactly 4, 8, 16; the next one, 32, exceeds n . $\square$

By Proposition 1, if for every $n \leq 30$ no minimum-degree-3 graph on n vertices contains C_4 , C_8 , or C_{16} is found, then no minimum-degree-3 counterexample on at most 30 vertices exists. A minimum-degree-3 graph requires $n \geq 4$, and for $n \leq 16$ the conjecture is the established baseline, which we reproduce (§4; the smallest orders $n = 4, 5$ are also covered by the independent CEGAR check below and trivially — K_4 contains C_4). It therefore suffices to settle $17 \leq n \leq 30$, which is the content of Theorem 1.

SAT Modulo Symmetries. SMS performs complete, isomorph-free generation of graphs satisfying given constraints: its modified CaDiCaL solver carries an in-search canonicity (minimality) propagator that prunes any partial adjacency matrix that cannot be the lexicographically minimal representative of its isomorphism class [11]. This is the key to scaling: the search considers essentially one graph per isomorphism class rather than enumerating all labelings.

Encoding. The minimum-degree constraint $\delta(G) \geq 3$ is encoded as a CNF cardinality constraint over the $\binom{n}{2}$ edge variables (we use SMS’s built-in sequential-counter encoding; see §4 for a totalizer cross-check). The constraint “no C_4 , C_8 , or C_{16} ” is enforced by the Glasgow subgraph solver [14] as a complete forbidden-subgraph propagator: whenever a (partial) graph contains one of these cycles as a (non-induced) subgraph, the propagator adds a clause excluding it. Forbidding the short cycle C_4 statically in CNF is cheap, but C_8 and C_{16} have $\Theta(n^8)$ and $\Theta(n^{16})$ potential occurrences, so a static encoding is infeasible; the propagator is what makes the search tractable.

Why not plain CEGAR-SAT. As a baseline we also implemented a self-contained counterexample-guided (CEGAR) SAT solver in Python (PySAT with CaDiCaL, a DFS power-of-two cycle detector, and a static adjacent-transposition lexicographic symmetry break). With only a partial symmetry break it re-discovers each forbidden cycle in many isomorphic copies: at $n = 17$ it accumulates roughly 85,000 refinement clauses and the per-instance time grows by about a factor of two per vertex, so it stalls at $n = 19$ (bound ≥ 20). SMS decides the same $n = 17$ instance in 2.9 seconds (Table 1). Complete symmetry breaking is precisely what removes this bottleneck; the two solvers agree wherever both finish ($n \leq 19$, §4).

3 Results

Table 1 reports, for each n from 17 to 30, that SMS finds *no* minimum-degree-3 graph avoiding all of C_4, C_8, C_{16} (solution count 0). All runs used a single CPU core.

Experimental setup. For each n , the minimum-degree-3 CNF is generated by PySMS’s `GraphEncodingBuilder` and the search is run by `smsg` with the invocation

```
smsg --vertices  $n$  --all-graphs --hide-graphs --forbidden-subgraph-file CYC --dimacs ENC
```

where `CYC` lists C_4, C_8, C_{16} (one per line, as $\langle k \rangle v_0 v_1 v_1 v_2 \dots$) and `ENC` is the generated DIMACS file. We use SMS (github.com/markkirch/sat-modulo-symmetries, version `v2.0.0-3-g464f12f`, commit `464f12f`) built with the Glasgow Subgraph Solver (commit `abd331a`) and the SMS-bundled CaDiCaL (`rel-2.1.2-38`) compiled with `g++ 12.2.0`. Each order ran on a single CPU core (a Modal `debian_slim`, Python 3.12.6 container). Wall times in Table 1 exclude container start-up.

n	count	wall time (s)
17	0	2.9
18	0	7.8
19	0	24.1
20	0	27.8
21	0	19.3
22	0	101.1
23	0	202.9
24	0	148.3
25	0	339.8
26	0	414.8
27	0	1000.3
28	0	1892.0
29	0	2342.8
30	0	6888.9

Table 1: SMS decision per order n : no minimum-degree-3 graph on n vertices avoids C_4, C_8, C_{16} . Together with the reproduced $n \leq 16$ baseline this proves Theorem 1.

The complete build and run scripts are in the accompanying repository; we recommend pinning the three commits above when reproducing.

4 Verification

A non-existence claim at $n = 30$ cannot be brute-forced (exhaustive generation of minimum-degree-3 graphs is infeasible beyond roughly $n = 13$), so we corroborate Theorem 1 with several independent checks. *No check produced a contradiction.*

1. **Ground-truth anchor.** At $n = 10$, forbidding only C_4 , SMS returns exactly 5 graphs, matching the independent count of the 5 C_4 -free minimum-degree-3 graphs on 10 vertices obtained with **nauty** (**geng+labelg**) [15].
2. **Baseline reproduction.** For $n = 6, \dots, 16$, forbidding all of C_4, C_8, C_{16} yields 0, reproducing the published baseline.
3. **Independent second solver.** Our CEGAR-SAT solver and SMS both return UNSAT for $n = 17, 18, 19$ (different solvers, symmetry breaking, and cycle handling).
4. **Encoding robustness.** Re-deciding with the totalizer cardinality encoding (a structurally different CNF) gives count 0 at $n = 17, 20, 22, 25$.
5. **Symmetry-method robustness.** Re-deciding with SMS’s colex minimality ordering gives count 0 at $n = 17, 20$; the (slower, experimental) colex variant timed out at $n = 22, 25$ (inconclusive, not contradictory).
6. **Positive controls.** Forbidding only C_4 (a weaker constraint) yields a graph at $n = 17, 20, 25, 30$, confirming the pipeline returns a solution when one exists.

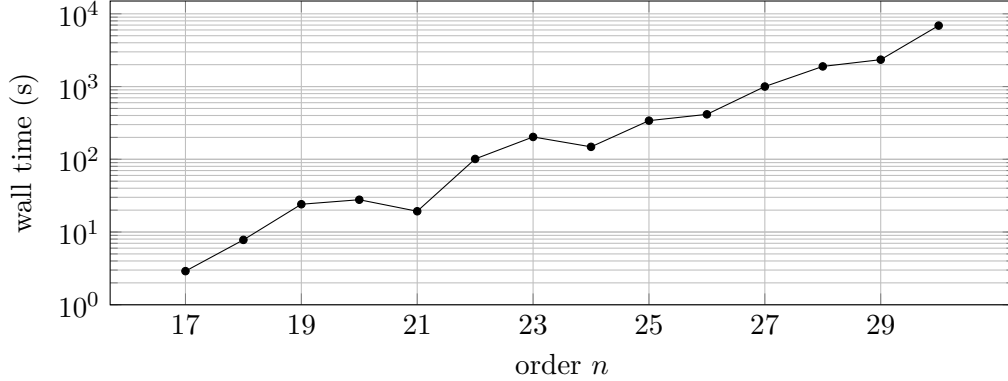


Figure 1: Single-core wall time of the SMS decision at each order n (log scale). Every point is UNSAT (no minimum-degree-3 graph on n vertices avoids C_4, C_8, C_{16}). Growth is roughly geometric, about a factor of 1.5–2 per added vertex.

Toward a machine-checked certificate. `msg` can emit an LRAT proof for UNSAT instances, but a generic LRAT checker cannot validate it against the minimum-degree-3 CNF alone, because the forbidden-cycle clauses are added by the propagator during search and are not RUP/RAT-derivable from that CNF (they exclude valid minimum-degree-3 graphs). A fully machine-checked certificate therefore requires the certified-SMS proof-logging machinery, in which the propagator justifies its own clauses; we leave this as the principal next step.

5 Discussion and limitations

Theorem 1 advances the general minimum-degree-3 frontier from 16 to 30 vertices (bound $17 \rightarrow 31$) and, as a byproduct, the cubic frontier from 29 to 30 (bound $30 \rightarrow 31$). Two caveats apply. First, this is a *computational* result: its soundness rests on SMS’s isomorph-free generation, the completeness of the Glasgow subgraph propagator, and the correctness of the min-degree encoding. The checks of §4 corroborate this composition — exactly, against `nauty` ground truth at $n = 10$, and against the established baseline for $n \leq 16$ — as is standard for computer-assisted enumeration; the natural further strengthening, for a nonexistence result of this kind, is an end-to-end machine-checked proof certificate (§6). Second, the result does not prove the conjecture; it only raises the lower bound on the order of a hypothetical counterexample.

6 Conclusion and future work

We have given the first SAT-based attack on the Erdős–Gyárfás conjecture. Casting the question as “does a minimum-degree-3 graph with no C_4 , C_8 , or C_{16} exist?” and discharging it with SAT Modulo Symmetries and the Glasgow subgraph propagator, we verified the conjecture for all minimum-degree-3 graphs on at most 30 vertices, raising the general lower bound on a counterexample from 17 to 31 (Theorem 1). The decisive ingredient is complete, in-search symmetry breaking: a conventional CEGAR encoding stalls near $n = 20$, whereas SMS reaches $n = 30$ in under two hours of single-core time.

Several directions remain. (i) *Push the frontier.* The runtimes (Figure 1) suggest $n = 31$ and a few further orders are within reach of a longer single run or modest parallelism; the present search was halted at $n = 30$. (ii) *A machine-checked certificate.* Replacing the verification checks

of §4 by an end-to-end, formally verified proof — via the certified-SMS proof-logging machinery, so that the forbidden-subgraph and canonicity propagators justify their own clauses — would put the result on the same footing as certified SAT resolutions of, e.g., the Pythagorean triples and Schur-number problems. (iii) *Exploit structure*. Carr’s result that a minimal counterexample is predominantly cubic [10] could be added as a propagator to prune the search and reach larger n . (iv) *Other questions*. The same SMS-plus-forbidden-subgraph template applies directly to related cycle-spectrum problems. Most importantly, an independent third-party reproduction would close the remaining gap between a corroborated computation and an established result.

Reproducibility

All code, the per- n data, and the verification scripts are available at <https://github.com/ArjunBalaji79/erdos-gyarfas-min-degree-3>.

Disclosure of generative AI use

In preparing this work the author used Anthropic’s Claude (the Claude Opus 4.8 model, via the Claude Code interface) to implement the SAT/SMS pipeline and the verification scripts, to configure and run the computational experiments, and to assist with drafting and the literature review. The author reviewed the generated code, outputs, and text, checked the results through the verification of §4, and takes full responsibility for the content and claims of this paper. No AI system is an author.

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